

A Specification-Traceable RTL and Verification Framework for a UCIe 2.0 Link-Training Control Subset

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Abstract—This work presents a specification-traceable register-transfer-level and verification framework for a bounded Universal Chiplet Interconnect Express 2.0-derived link-training control problem. The synthesizable design combines a nine-state hierarchical controller, bounded sideband retry and response matching, a 16-lane 23-bit linear-feedback shift-register measurement engine, retained training-error classification, and a compact field-programmable gate array control/status-register wrapper. Production verification disables the abstract training bypass and independently predicts message order, accepted-sample lane state, strict error-threshold decisions, abort cause, and state advancement. Five fixed seeds execute 180 trials covering nine nominal, retry, malformed-response, pattern-failure, reset, fatal, and timeout classes; accepted logs contain no verification errors or fatals, and 398 independent predictor checks complete. On a Cyclone 10 LP device, the wide diagnostic top fails to fit at the 151-pin package limit, whereas the byte-register wrapper fits with 824 logic elements, 505 registers, and 119 pins. Internal 80-MHz timing reports positive setup and hold slack. The evidence supports a reproducible academic digital-control case study, not a complete physical layer, electrical link, interoperability result, or standards-compliance claim.

Index Terms—chiplets, FPGA, link training, UCIe, UVM.

I. INTRODUCTION

Chiplet-based systems partition functions across dies and reconnect them within a package. This decomposition can improve reuse and allow each function to use an appropriate process, but it moves system correctness into the die-to-die interface and package boundary [1], [2]. UCIe defines an open layered interface intended to support such heterogeneous integration [3]. Within its physical-layer control, the Link Training State Machine (LTSM) coordinates reset, sideband initialization, mainband initialization and training, active operation, retraining, error, and low-power behavior [4].

An academic RTL model of this control is easy to overstate. Listing standard states does not implement their physical procedures; a passing nominal simulation does not test message ordering under backpressure; and an FPGA synthesis report does not establish a working UCIe link. Reproducible negative evidence is equally important: failed fits, malformed responses, pattern failure, abort timing, and retained error behavior define the boundary of what has actually been shown.

This paper presents a source-available, specification-traceable RTL and verification framework for a UCIe 2.0-

derived digital-control subset. The synthesizable design combines a nine-state hierarchical controller, a bounded retrying sideband sequencer, a 16-lane 23-bit LFSR measurement engine, retained TRAINERROR classification, and a compact FPGA CSR wrapper. Its detailed case study is the production DATATRainingCENTER1 path, in which START completion admits pattern measurement, pattern failure retries without END, and a matching END response gates advancement.

The paper makes three bounded contributions:

- 1) a modular controller architecture that integrates sideband transaction, accepted-sample LFSR measurement, error retention, and compact observability without presenting the result as a complete UCIe PHY;
- 2) an interface-derived UVM predictor and SVA layer that independently check message order, per-lane LFSR state, strict threshold semantics, abort cause, and state advancement; and
- 3) a reproducible evidence chain comprising directed suites, preserved multi-seed campaigns, 180 production integrated trials across nine scenario classes, and qualified Quartus results that retain both a failed wide-top fit and a successful 119-pin CSR-wrapper fit.

Section II defines the implementation boundary and RTL. Section III describes prediction, assertions, and scenario accounting. Section IV reports simulation and FPGA evidence. Sections V and VI position the contribution and state its limitations.

II. CONTROLLER SCOPE AND ARCHITECTURE

A. Boundary of the Model

UCIe separates protocol, die-to-die adapter, and physical-layer concerns, with the sideband supporting training, register access, and diagnostics [3], [5]. This work addresses a digital-control slice within that broader stack. As Fig. 1 shows, the environment provides transaction-level sideband responses, training samples, and abstract phase completions. The controller returns state/status, sideband requests, training patterns, and retained diagnostics. Analog signaling, the package channel, mainband data transfer, complete adapter/protocol operation, and compliance are deliberately outside the boundary.

The modeled state and substate names follow UCIe Specification Revision 2.0, Version 1.0 [4]. Most MBINIT, MBTRAIN, and PHYRETRAIN operations are represented by an external phase_done_i condition. One training operation, DATATRainingCENTER1, is implemented through the sideband

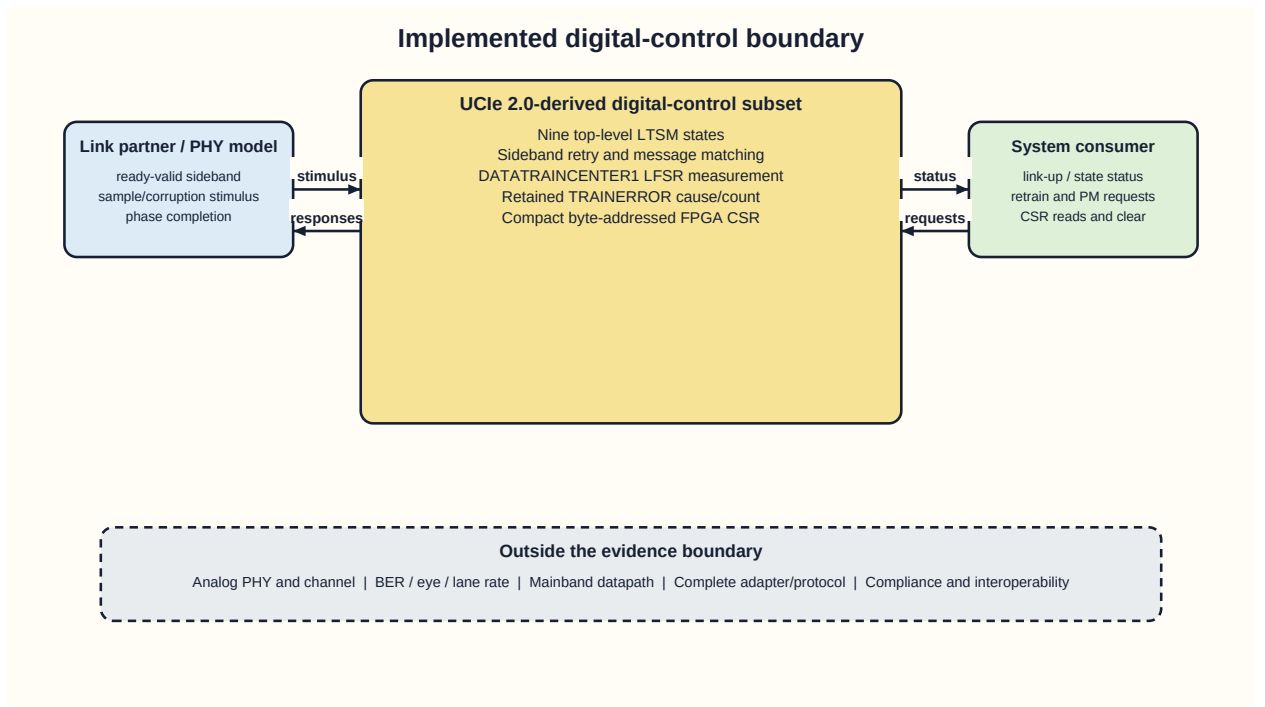


Fig. 1. Implemented digital-control boundary. Gray/dashed content is not implemented or inferred from the simulation and FPGA evidence.

and LFSR engines. The distinction prevents a control-flow model from being mistaken for a complete PHY.

B. Module Decomposition

Fig. 2 shows the synthesizable hierarchy. The top-level `ucie_ltсм` owns state/substate registers, a residency timer, routing, and engine coordination. Three engines isolate operations whose temporal contracts differ from ordinary state progression.

The sideband sequencer has IDLE, SEND, WAIT, and ERROR states. It holds the request under ready/valid backpressure, accepts only the expected response, retries a bounded number of response timeouts, and exposes protocol error and abort status. The current message enumeration includes NOP, SBINIT-DONE request/response, and DATACENTER1 START/END request/response. It is not a complete sideband packet layer.

The LFSR engine maintains 16 independent 23-bit states. Eight nonzero seeds are repeated over the lanes, and the feedback polynomial is $x^{23} + x^{21} + x^{16} + x^8 + x^5 + x^2 + 1$. State and sample count advance only when `train_rx_valid_i` accepts a sample. Mismatches accumulate in a 16-bit saturating counter, and pass is asserted only when the final count is strictly less than the programmed threshold; equality fails. Fig. 3 summarizes the datapath and its independent reference boundary.

The error manager accepts one eligible state-timeout, sideband-protocol, or local-fatal event. Simultaneous events use the fixed priority timeout, sideband, then fatal. A pending event holds the TRAINERROR request until an acknowledgment or manager timeout permits entry. Cause and a saturating 16-bit event count remain stable in TRAINERROR.

A software clear is accepted only when no event is pending and the controller is outside TRAINERROR.

C. Hierarchical LTSM Control

The package enumerates nine top states: RESET, SBINIT, MBINIT, MBTRAIN, LINKINIT, ACTIVE, PHYRETRAIN, TRAINERROR, and L1/L2. MBINIT contains six substates and MBTRAIN contains thirteen. The nominal route progresses through initialization and training to ACTIVE. From ACTIVE, requests can select one of three retraining targets or enter the modeled low-power state; exit conditions return to ACTIVE. Eligible timeout, protocol, or fatal conditions route through the retained error manager to TRAINERROR, as shown in Fig. 4.

One timer measures state/substate residency. It restarts on a state or substate change and can be stalled by the environment. Default parameters correspond to an 80-MHz controller clock, a 4-ms reset interval, an 8-ms residency timeout, a 256-cycle sideband response timeout, one retry, and 4096 training samples. Testbenches scale selected values for runtime while retaining the ordering properties under test.

D. Integrated DATATRANCENTER1 Protocol

Production mode fixes the abstract CENTER1 bypass to zero. On entering DATATRANCENTER1, the controller orders the sideband engine to send START and wait for its matching response. Only then may the LFSR engine begin. A failed pattern re-enters the pattern phase without emitting END; a passing pattern starts the END exchange. Advancement to DATATRANCENTER1 requires the matching END response. A mismatch, exhausted sideband retry, fatal

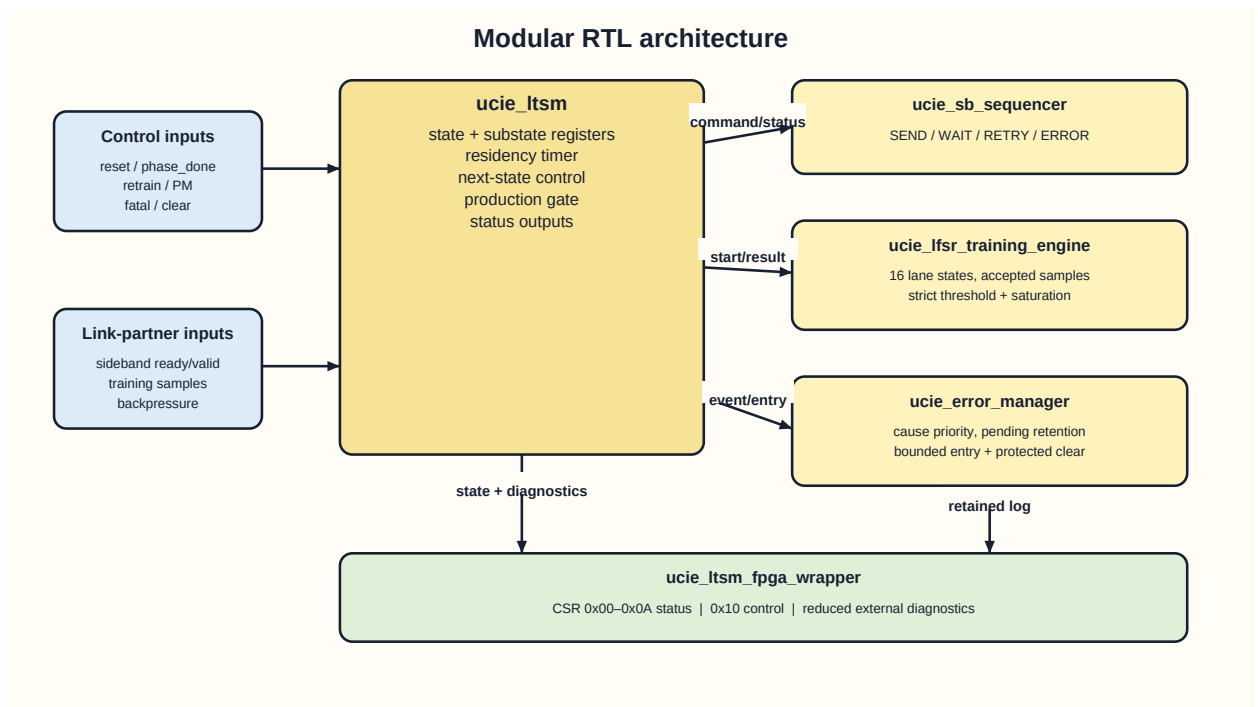


Fig. 2. Modular RTL architecture. The wrapper does not alter the controller; it serializes internal status through a compact CSR interface.

event, state timeout, or reset aborts the sequence along its corresponding control path.

This ordering, illustrated in Fig. 5, creates two explicit commit barriers: the START response admits pattern activity, and the END response admits state advancement. Separating the sideband and pattern engines keeps retry time from advancing training state and keeps receive gaps from consuming the sideband timeout counter.

E. Retained Diagnostics and FPGA CSR

The compact wrapper preserves the controller interface internally but replaces wide state/error/training diagnostics with byte reads. Read addresses 0x00, 0x01, and 0x02 expose top, MBINIT, and MBTRAIN state; 0x03 exposes status bits; 0x04 exposes error cause; 0x05–0x06 and 0x07–0x08 expose retained-event and training-error counts; 0x09 returns version 0x04; and 0x0A exposes the integrated training phase. Writing bit 0 at 0x10 requests a protected log clear. Undefined reads return zero.

Fig. 6 connects the retained log, entry handshake, recovery route, CSR view, and assertion boundary. This observability choice is not only documentation convenience: Sec. IV shows that exporting wide diagnostics prevents the chosen FPGA from fitting, whereas the CSR wrapper fits.

III. SPECIFICATION-TRACEABLE VERIFICATION

The verification structure in Fig. 7 separates stimulus, observation, prediction, assertions, and coverage accounting. The separation is important for training logic: a scoreboard that predicts from the DUT's internal LFSR state or phase register can agree with the same defect it is meant to detect.

A. Directed and Randomized Layers

Directed tests isolate handshake and boundary semantics before integration. The sideband suite covers successful exchange, timeout/retry, retry exhaustion, response mismatch, and abort. LFSR tests cover nonzero seeds, the implemented polynomial, receive gaps, strict threshold equality, abort, the default 4096-sample configuration, and 16-bit saturation. Error-manager tests cover cause priority, retention, immediate/acknowledged/timed entry, protected clear, and count saturation. Wrapper tests read version, state, status, training phase, cause/count, allowed and ignored clear operations, and invalid addresses.

Nine deterministic UVM tests retain top-level state and transition checks. Separate seeded campaigns stress sideband, training, and recovery behavior. The production integrated test disables the abstract CENTER1 bypass and uses nine scenario classes: nominal completion; pattern failure followed by retry and pass; START retry; malformed START response; END retry; malformed END response; and reset, fatal, or state-timeout aborts. Backpressure, response latency, sample gaps, corruption, threshold, and abort phase vary within these classes.

B. Independent Predictor

The predictor observes transmitted requests, received responses, accepted training samples, and public controller/error status. It maintains its own 23-bit state for each of 16 lanes, uses the specified project seed table and polynomial, advances only when a receive sample is accepted, and computes the expected mismatch count independently. It also maintains a protocol phase: START must close before pattern activity, END must not appear before a passing pattern, and

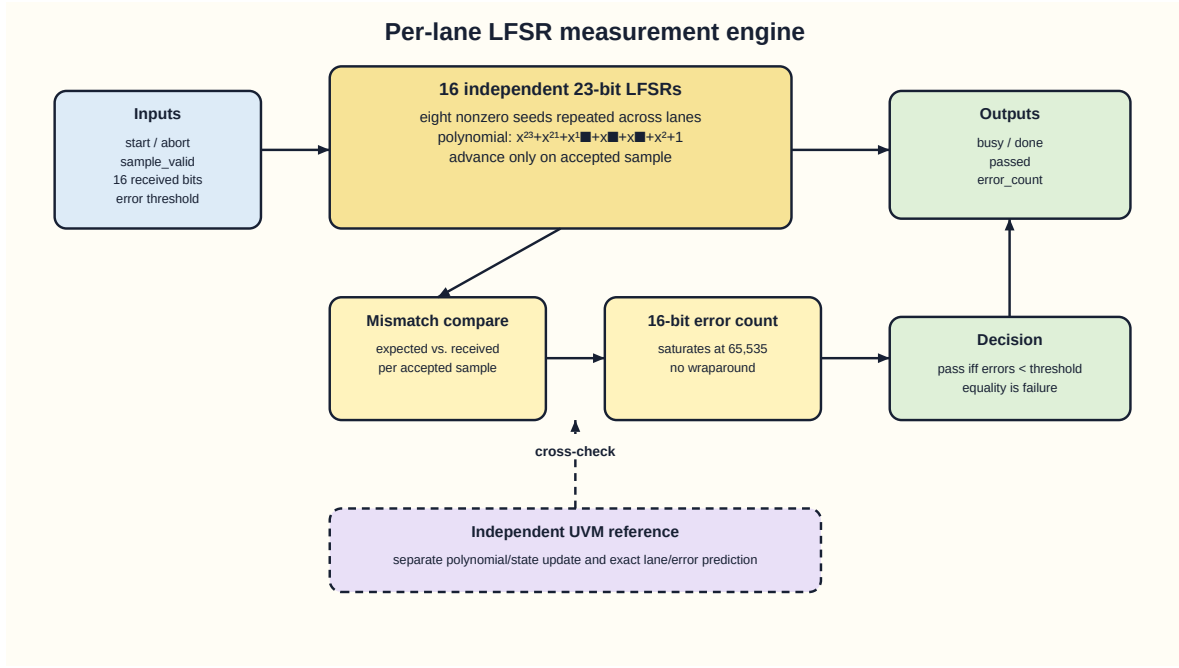


Fig. 3. LFSR measurement engine. Receive gaps freeze both the sample count and per-lane expected state.

DATATRainingVREF must not be reached until END closes. Abort scenarios check both clean cessation and retained cause attribution.

C. Assertions and Coverage Boundary

Concurrent assertions check that `link_up` is exclusive to ACTIVE, timeout routes to TRAINERROR, reset cannot jump directly to ACTIVE, and states remain known. Integrated assertions restrict pattern activity to DATATRainingCENTER1/PATTERN, constrain requests to their phase, require an END response before production-mode advance, check the strict error threshold, and hold LFSR progress when no sample is accepted. Recovery properties cover pending request retention, bounded TRAINERROR entry, pending clear, log stability, and ignored clear in TRAINERROR.

Explicit counters record scenario, phase, gap, corruption, threshold, retry, abort, and selected cross combinations. They demonstrate that planned classes were sampled, but are not reported as a standard functional-coverage percentage because no UCDB was produced under the available Starter license.

IV. EVALUATION

The evaluation asks two bounded questions: whether the production controller preserves the required START–PATTERN–END ordering under randomized delays and faults, and whether the same RTL is implementable through a practical FPGA observability boundary. It does not measure link throughput, latency, BER, or electrical behavior.

A. Environment and Reproducibility

Simulation used Questa Intel FPGA Starter Edition 2023.3. The committed PowerShell and Questa scripts compile the

RTL, UVM packages, and assertions, then run fixed seed lists. The integrated testbench drives a 100-MHz simulation clock, reduces the training sample count to eight accepted samples, and fixes ALLOW_ABSTRACT_DATATRaining_BYPASS to zero. These changes shorten the experiment without changing the message-order, accepted-sample, threshold, or abort checks. The RTL default remains 4096 accepted samples.

FPGA mapping, fitting, and timing analysis used Quartus Prime Lite 23.1std.1, targeting a Cyclone 10 LP 10CL025YU256C8G. The SDC applies a 12.5-ns (80-MHz) clock constraint and an asynchronous-reset false path. It does not assign package locations or external input/output delays; timing values are therefore internal-clock evidence, not board timing closure.

B. Production Integrated-Training Campaign

Five seeds each execute 36 trials. Every seed first guarantees all nine scenario classes and then adds seeded variations in response latency, backpressure, sample gaps, corruption, threshold, and abort phase. Table I reports the accepted per-seed summaries. A request need not receive a response in malformed-response, retry, or abort cases, and END is intentionally absent when the pattern does not pass. The resulting count asymmetry is therefore expected, not packet loss.

The scenario-hit totals were 18 nominal passes, 20 failed-pattern retry/pass cases, 17 START retries, 16 malformed START responses, 23 END retries, 19 malformed END responses, 20 reset aborts, 18 fatal aborts, and 29 state-timeout aborts. Thus all planned classes were observed, while the predictor performed 398 independent order/value/cause checks. The accepted logs report zero UVM errors and zero UVM fatalities, and the regression summary reports no assertion failures.

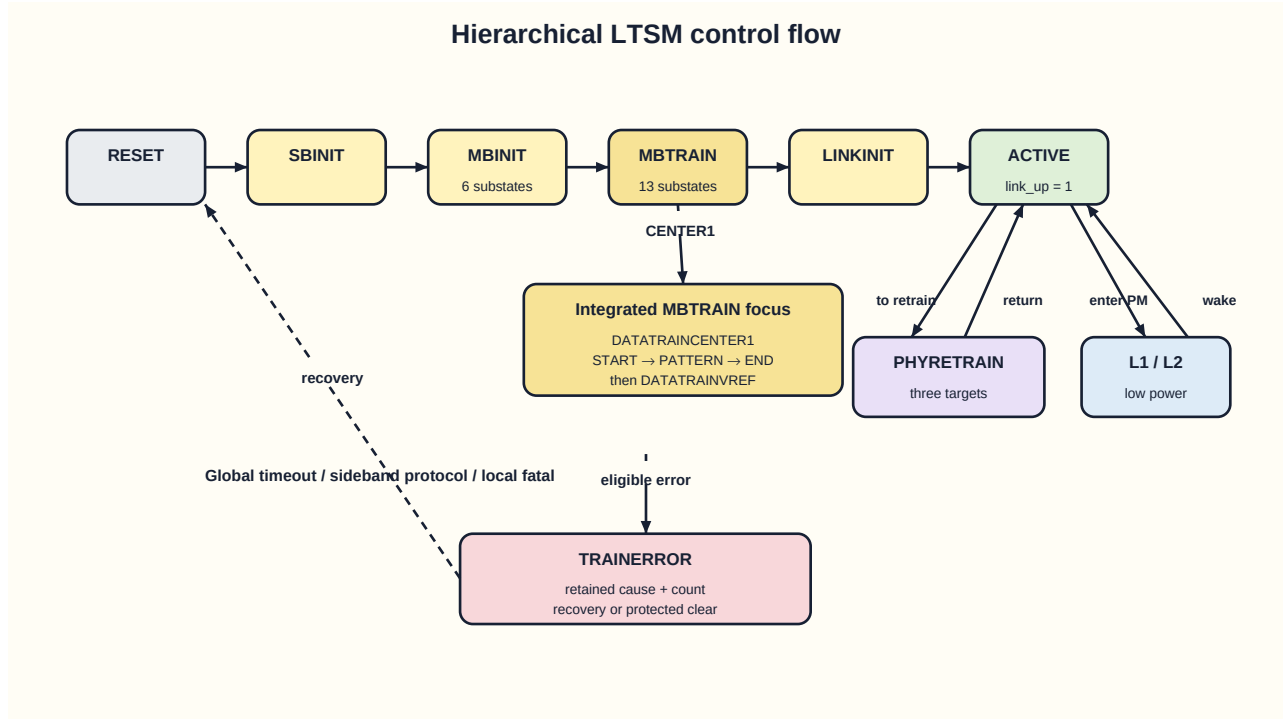


Fig. 4. Hierarchical controller flow. Only DATATRAINCENTER1 is expanded here; other physical operations remain phase-completion abstractions.

TABLE I
PRODUCTION INTEGRATED-TRAINING RESULTS WITH BYPASS DISABLED

Seed	Trials	START Req.	START Resp.	END Req.	END Resp.	Retry	Prot. err.	Pattern Pass	Pattern Fail	Abort Reset	Abort Fatal	Checks T/O	Checks
1701	36	36	30	25	17	8	6	25	4	8	1	4	83
1802	36	36	28	23	12	8	11	23	2	3	3	7	76
1903	36	32	28	21	14	5	6	23	4	3	6	7	78
2004	36	34	26	24	18	8	6	23	6	2	3	7	80
2105	36	35	28	27	17	11	6	27	4	4	5	4	81
Total	180	173	140	120	78	40	35	121	20	20	18	29	398

These observations establish consistency for the committed stimuli; they are not an exhaustive proof.

C. Preserved Regression Gate

The integrated campaign was added without discarding earlier evidence. The release gate also ran nine deterministic UVM tests, 200 sideband trials over seeds 101–505, 160 LFSR-training trials over seeds 701–1105, and 180 recovery trials over seeds 1201–1605. Module-directed suites separately exercise the controller, sideband sequencer, LFSR engine, error manager, and wrapper CSR. All accepted runs completed without UVM errors/fatals or reported assertion failures. An ad-hoc ignored debug log is not part of this evidence set.

D. FPGA Mapping, Fit, and Timing

Table II preserves both the positive and negative implementation results. The wide controller top maps to 763 logic elements and 505 registers, but demands all 151 device pins; its fitter run fails. Timing numbers emitted alongside that failed fit are not used as successful-fit evidence.

The CSR wrapper fits successfully using 824 of 24,624 logic elements (3%), 505 registers, and 119 of 151 pins (79%), with no virtual pins. Across the committed timing corners, the worst setup and hold slacks are +1.013 and +0.179 ns, respectively, and reported negative slack is zero. The wrapper adds a small amount of decode logic while reducing external pin demand by 32, demonstrating why serializing diagnostics is necessary for this device. Because external I/O constraints and physical pin locations are absent, the result supports internal synthesizability and fit only.

E. Threats to Validity

The pseudo-random campaign uses explicit seeded \$urandom_range choices and sampled counters because the Starter license does not support the native class-randomization and UCDB coverage flow required by the original plan. No native functional-coverage percentage or code coverage is claimed. The independent predictor reduces common-mode risk, but design and verification were developed in the same project. Reduced training counts

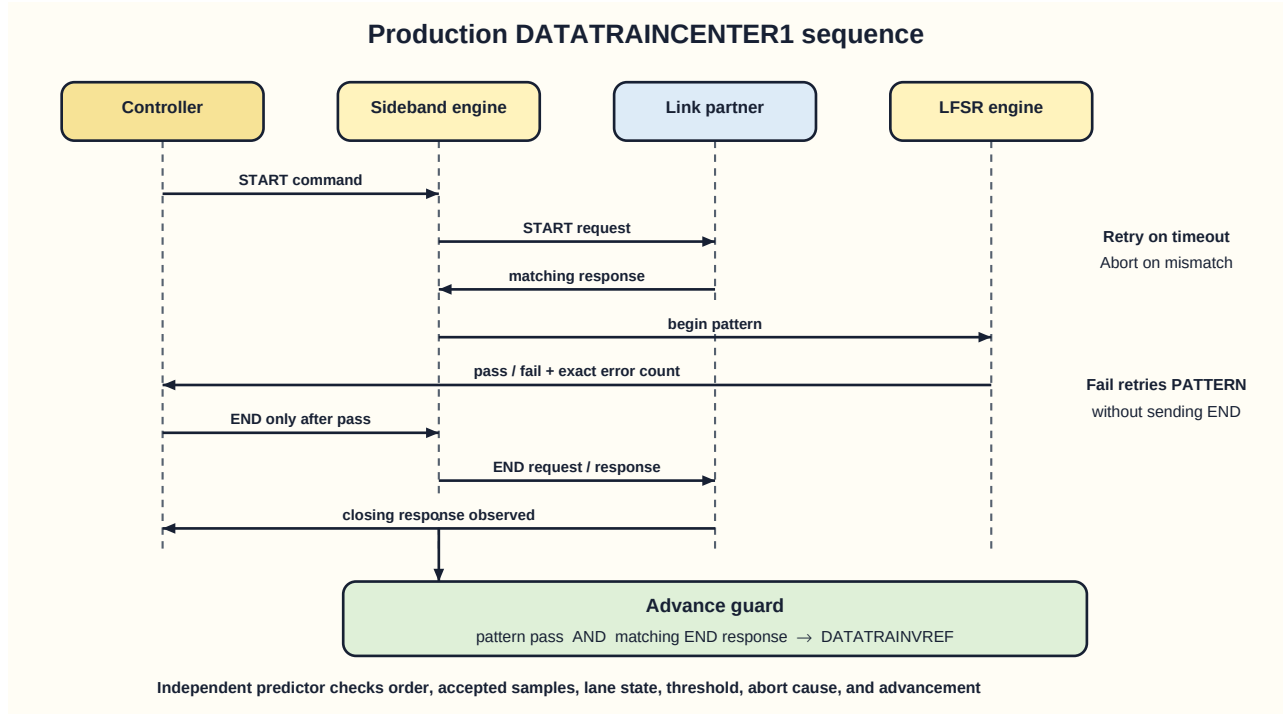


Fig. 5. Integrated CENTER1 transaction order. The independent predictor checks the same interface events without reading the DUT’s private LFSR state.

TABLE II
CYCLONE 10 LP IMPLEMENTATION SUMMARY

Metric	Core map	Core fit	Wrapper fit
Status	success	failure	success
Logic elements	763	764	824
Registers	505	505	505
Pins	151	151	119
Setup slack (ns)	–	–	+1.013
Hold slack (ns)	–	–	+0.179

accelerate simulation; they do not test full-duration drift. Finally, FPGA synthesis cannot substitute for a complete two-die model, board experiment, ASIC PPA, or electrical characterization.

V. RELATED WORK AND POSITIONING

Chiplet integration combines smaller dies through advanced packaging to improve reuse, yield options, and technology specialization, while introducing new interconnect, test, thermal, and integration challenges [1], [2]. UCIE addresses the interoperability problem with a layered die-to-die standard. Sharma *et al.* describe the architecture, circuit, channel, and packaging rationale behind UCIE 1.0 [3]; later work examines memory/storage applications [6] and circuit/package scaling toward three-dimensional integration [5]. Those studies establish UCIE’s system and physical context. This work does not reproduce their bandwidth, energy, or channel claims; it targets the digital training-control evidence boundary.

Physical UCIE studies further illustrate why that boundary matters. Thermal and signal-integrity analysis of UCIE-A channels [7] and FPGA-based high-speed interface verification

platforms [8] operate with channel, transceiver, or measurement evidence that is absent here. Consequently, successful RTL simulation and FPGA fitting cannot imply BER, electrical margin, or interoperability.

Verification practice commonly combines directed bring-up, constrained-random exploration, scoreboards, assertions, and coverage. IEEE Std 1800.2 defines the UVM framework [9]. Vamsikrishna *et al.* describe a unified testbench supporting directed, random, and pseudo-random operation selection, including a UCIE-oriented example [10]. Intel and Cadence report PHY-model simulation interoperability in which sideband transactions and high-level LTSM state stepping coordinate vector traffic [11]. These public accounts demonstrate mature methodology and commercial model use, but do not provide a directly comparable open RTL controller, independent lane-LFSR/message-order predictor, retained-error CSR, and fixed-seed evidence chain.

The contribution here is therefore the auditable combination rather than any individual technique: a bounded UCIE 2.0-derived RTL control subset, an interface-derived predictor and SVA layer, explicit negative scenarios, and qualified FPGA evidence with a preserved failed-fit result. No “first” claim is made; public literature and proprietary implementations continue to evolve, including the newer UCIE 3.0 specification [12].

VI. DISCUSSION

A. What the Evidence Supports

The strongest result is control-path consistency across independently predicted message order, per-lane pattern state, threshold decisions, and retained abort cause. The tests exercise not only nominal completion but also absence properties:

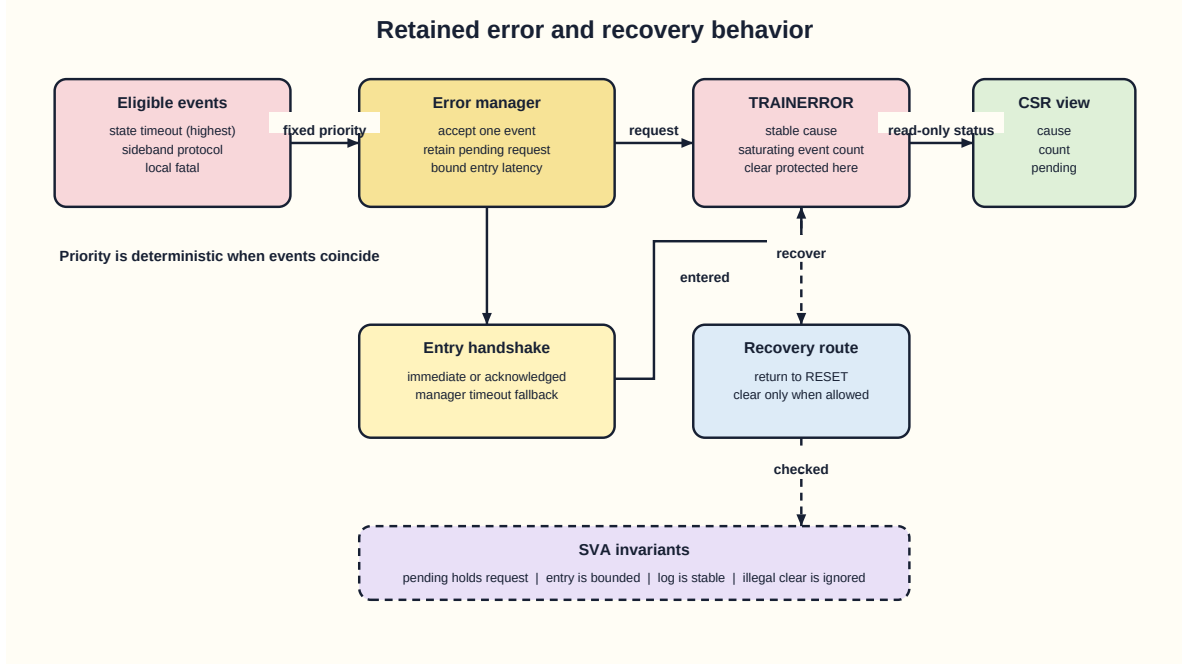


Fig. 6. Retained error and recovery flow. Clear is ignored while an event is pending or the state is TRAINERROR.

END must not be sent after a failed pattern, DATATRainingVREF must not be entered before END closes, and illegal clear must not erase a retained TRAINERROR record. These checks are more informative for the modeled block than a link-level throughput number that the design cannot produce.

The FPGA result also shows a practical architectural trade-off. Exposing every diagnostic directly consumes the complete package-pin budget and fails to fit, although logic utilization is low. A byte CSR raises logic use from the mapped 763-element core to a fitted 824-element wrapper while reducing pin demand by 32. This is not a performance optimization; it is an observability boundary that makes the controller implementable on the selected package.

B. What the Evidence Does Not Support

The controller implements one detailed training operation while delegating most other substates to `phase_done_i`. The project therefore cannot claim a complete physical-layer controller. It has no analog front end, package channel, main-band datapath, protocol adapter, two-die interoperability setup, or official compliance suite. The Quartus constraint set has no board pin locations or external I/O delays, and no hardware was measured. There is no ASIC synthesis, place-and-route, power analysis, CDC/RDC signoff, formal proof, or native code coverage.

The fixed-seed campaign improves reproducibility but samples a finite behavioral space. Explicit bin counters confirm that planned classes were exercised; they do not establish standard coverage closure. A stronger evaluation would run a licensed native randomization/coverage flow, publish functional and code coverage with justified exclusions, and use mutation or fault injection to measure checker sensitivity.

C. Next Experiments

Three extensions would most improve the scientific claim. First, replace more phase abstractions with independently checked MBINIT/MBTRAIN procedures and a two-end behavioral partner. Second, compare controller-only, integrated-engine, and CSR-wrapper configurations under identical FPGA or ASIC constraints. Third, add physical credibility through either a fully constrained board build and hardware experiment or a reproducible ASIC flow with library, PVT, constraint, area, and power methodology. Until then, the work is best described as an academic digital-control and verification framework.

VII. CONCLUSION

This work develops and audits a UCIE 2.0-derived digital-control subset rather than a complete link. The RTL joins hierarchical LTSM sequencing, bounded sideband transactions, per-lane LFSR measurement, retained error attribution, and a compact FPGA CSR. Production-mode verification keeps the abstract DATATRainingCENTER1 bypass disabled and uses an independent message-order/LFSR predictor, assertions, directed corner tests, and fixed-seed negative scenarios. Across 180 integrated trials, all nine planned scenario classes were observed with 398 predictor checks and no accepted-log UVM errors or fatals. Earlier sideband, training, recovery, deterministic, and directed suites also remain in the regression gate.

On the selected Cyclone 10 LP, the wide diagnostic top fails to fit at 151 pins, while the CSR wrapper fits with 824 logic elements, 505 registers, and 119 pins. Its internal 80-MHz timing reports positive setup and hold slack, but the absence of board I/O constraints prevents a board-closure

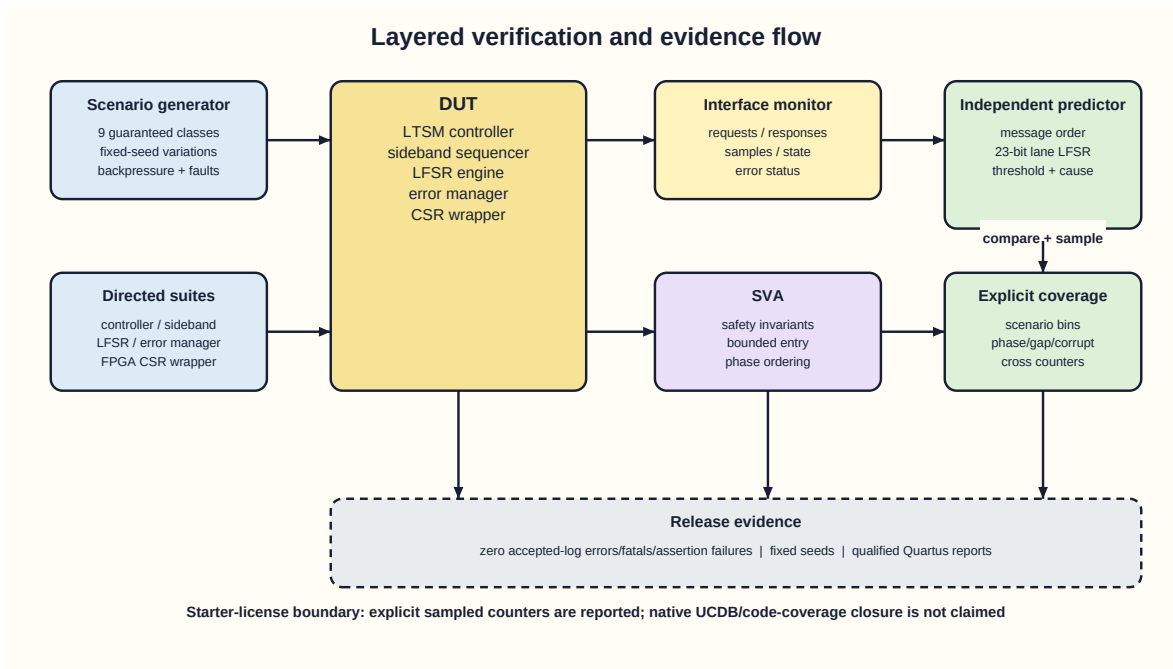


Fig. 7. Verification architecture. The predictor reconstructs message order, per-lane LFSR state, thresholds, and error cause from interface observations.

claim. The result is therefore a reproducible academic control/verification artifact. Complete training procedures, native coverage closure, hardware or ASIC evidence, and electrical/interoperability validation remain future work.

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The authors must replace this draft text with complete funding, institutional, conflict-of-interest, and contributor acknowledgments. OpenAI Codex was used under author direction to assist with repository auditing, evidence organization, manuscript drafting, and generation of original project figures. The authors are responsible for independently checking every claim, citation, figure, result, and disclosure and for approving the final submitted text.

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